

Can the on-structure sensors predict inclination?

A chronological NeuralProphet experiment on same-time estimation, one-week forecasting and missing-hour reconstruction

Study report · `studies/04_neuralprophet_inclination_prediction/`

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1 Introduction

Study 01 read the raw archive, decided which of the recorded values are measurements, and exported the verdict-aware archive that every later study reads. Its final inclination product —

compensated for temperature, anchored once across the whole record, cleaned of impulsive noise, and carrying a flag that marks every value it replaced by interpolation — is the only inclination series used here. The intermediate versions of that channel are not used, and no value the flag marks as an interpolation is treated as a measurement. This study inherits that verdict rather than revisiting it.

The subject is one sensor at one place: the inclinometer at station 02, on the medieval wall at Porta di Sant’Ubaldo. The question is operational rather than diagnostic. Given the record as it actually stands, can the next value of the inclination be predicted — from the inclination’s own history, from the other variables the instrument package on the structure measures beside it, or from external environmental proxies?

The record is not taken whole. The archive contains a 271-day outage that begins on 23 September 2022 and whose last missing day is 20 June 2023, across which nothing at all about the structure’s behaviour was recorded. This study works only on what follows: the stretch running from the resumption of recording in June 2023 to the end of the archive. The choice has two reasons. That stretch is the part of the record closest to the present, and therefore the part a monitoring system deployed today would actually have to work with. It is also the only part over which the complete instrument package is available, since the wall-temperature probe and the pyranometer were installed on 21 February 2025 and record nothing before that date.

Choosing that window does not buy a continuous series. Four further whole-day outages fall inside it — 40 days from 5 May 2024, 42 days from 28 August 2024, 17 days from 26 September 2025, and 103 days from 7 March 2026 — and beneath them lies a population of shorter dropouts whose size is not yet known. Over the 27,769 hourly slots the window spans, the inclinometer returns an accepted value in 76.2% of them, and air temperature and relative humidity in 78.4%. The wall-temperature probe and the pyranometer reach only 17.7% and 18.9%: they exist for less than half the window to begin with, and they are interrupted inside the part where they do exist. What none of those numbers says is how the missing time is *distributed*: whether it arrives as isolated absent samples, as hours, or as days. Study 01 recorded the omission explicitly, noting that the anatomy of the gaps is a natural extension of it and was not attempted in it. Establishing that anatomy is the first task of this study, because the shape of the missing data decides what kind of statistical problem filling it is: recovering a scattered sample from its immediate neighbours is a different proposition from reconstructing a season.

The study therefore proceeds in four movements. It first looks at the window whole, plotting every variable the on-structure package records over it on one shared clock, with the gaps left as gaps and nothing interpolated before drawing. It then inventories those gaps, classifying the missing time by duration so that the scattered absences are separated from the outages. It then weighs the ways each class might be filled, and what it would cost to be wrong. Only then does it put the prediction question: what the next inclination value can be inferred from, using the on-structure measurements.

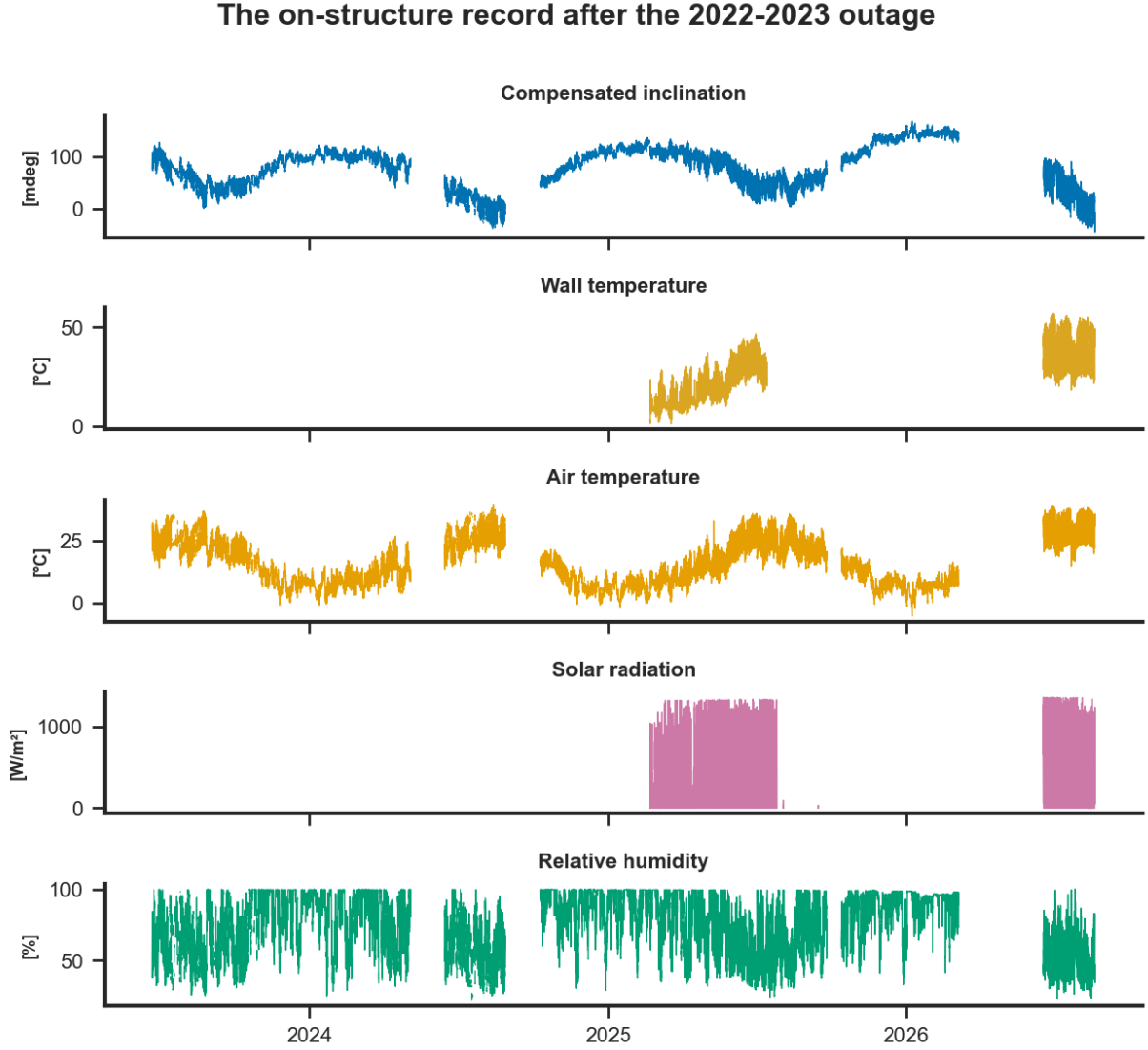


Figure 1: Every variable the instrument package on the wall records, over the window this study works on, on one shared clock. Nothing is interpolated and nothing is dropped before drawing: a gap in the record is a gap in the figure. Inclination, air temperature and relative humidity run from the resumption of recording in June 2023; wall temperature and solar radiation begin only with the package installed on 21 February 2025, and are themselves interrupted after it. The package also records its own supply voltage, which is instrument housekeeping rather than a measurement of the wall or of its environment, and is not drawn here.

2 The record this study works on

The window runs from 21 June 2023 to 20 August 2026 and spans 83,305 slots on the twenty-minute grid at which the acquisition system writes. That grid, rather than an hourly aggregate, is the one every result below is stated on, because it is the rate at which a deployed system receives a new reading and therefore the rate at which it must decide whether that reading is the expected one.

Table 1 gives what each channel returns over the window. The inclinometer delivers an accepted value in 75.5% of the slots, air temperature and relative humidity in 77.8% of them. Wall temperature and solar radiation reach 17.6% and 18.8%: the instruments that record them were installed on 21 February 2025 and so do not exist for the first three fifths of the window at all. That is the whole of the reason those two channels are excluded from every model in this study. Requiring them would not cost a percentage point of coverage but three fifths of the record, and every result would then have to be reported twice, on two windows that cannot be compared.

They remain in Figure 1 because what the package records is part of what this study has to describe, and a reader is owed the sight of the channels it set aside.

Table 1: Accepted values and coverage per channel over the window, on the twenty-minute grid. Read from `NP_01_window_coverage.csv`.

Channel	Accepted slots	Coverage
<code>inc_comp_cleaned</code>	62,931	75.5%
<code>twall_str</code>	14,632	17.6%
<code>tair_str</code>	64,830	77.8%
<code>sr_str</code>	15,682	18.8%
<code>rh_str</code>	64,830	77.8%

Air temperature and relative humidity cost this study nothing. Their missingness is nested inside the inclinometer’s: every slot in which the inclination is accepted is a slot in which both are accepted too. Requiring all three therefore leaves exactly the same rows, in exactly the same contiguous runs, as requiring the inclination alone. That is a property of this record and not a general one, and it is the reason the predictor set could be chosen on physical grounds rather than by trading coverage against information.

3 The anatomy of what is missing

A coverage percentage says how much time is missing. It does not say how that time is shaped, and the shape is what decides whether filling it is interpolation or reconstruction. Over the window the inclination record is interrupted 1,647 times, for a total of 6,791 hours.

Those interruptions are distributed with extreme asymmetry, and in two directions at once. Four gaps longer than seven days — 40 days from 5 May 2024, 42 days from 28 August 2024, 17 days from 26 September 2025 and 103 days from 7 March 2026 — carry 72.8% of all the missing time between them, while being 0.2% of the gaps. At the other end, 1,280 gaps of an hour or less carry only 8.6% of the missing time while being 77.7% of the gaps. Table 2 gives the full classification.

Table 2: The missing time by gap duration: how many gaps of each class, and how many hours they carry between them. Read from `NP_02_gap_inventory.csv`.

Gap class	Gaps	Hours
>7d	4	4,943.7
1-6h	315	759.0
<=1h	1,280	583.0
6-24h	48	505.7

The second number matters more than the first, and it is the one a coverage figure hides. The four long outages are conspicuous, and no honest method would try to reconstruct 103 days of a structure’s behaviour from its neighbours. It is the short dropouts that determine what can be modelled, because they are what breaks *contiguity*, and contiguity is the currency an autoregressive window spends. A model asked for 24 hours of history before each prediction cannot use a run of data shorter than that, however complete the run is.

Table 3 prices that directly. The record divides into 1,647 contiguous runs whose median length is 1.7 hours. Requiring 24 hours of history and forecasting 24 hours ahead leaves 101 of those runs usable and 15,000 prediction windows; requiring 48 hours of history for the same

forecast leaves 58 runs and 9,092 windows. The autoregressive window used in Section 8 is chosen from this table rather than from habit.

Table 3: What each autoregressive window costs in usable data. A contiguous run survives only if it is longer than the lag plus the forecast it must support. Read from `NP_03_segment_survival.csv`.

Lag (h)	Forecast (h)	Runs	Surviving	Windows
4	8	1,647	476	39,411
4	24	1,647	186	23,978
8	8	1,647	403	34,092
8	24	1,647	175	21,810
12	8	1,647	320	29,657
12	24	1,647	157	19,793
24	8	1,647	175	21,810
24	24	1,647	101	15,000
48	8	1,647	91	12,717
48	24	1,647	58	9,092

How the missing time is shaped

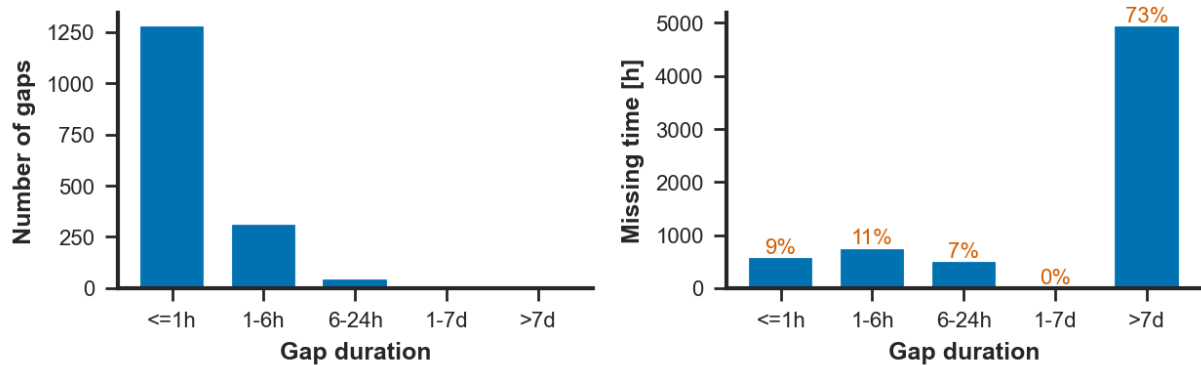


Figure 2: The anatomy of the missing time. The gaps are counted by duration class and weighted by the time each class carries, which is what separates the many short dropouts from the few long outages.

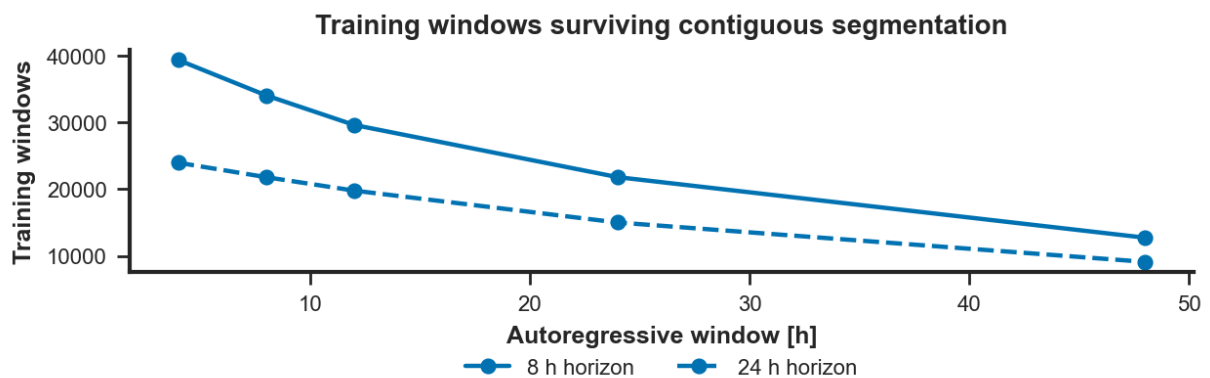


Figure 3: Contiguous runs surviving each autoregressive window, and the prediction windows they yield. The cost of a longer memory on a fragmented record is not linear: it falls on the number of runs that qualify at all.

4 Method

Each subsection below states one decision, the evidence that forced it, and what would have gone wrong under the alternative. They are given before any result that depends on them.

4.1 Why the level is decomposed but never scored

The inclination level has a lag-one autocorrelation of 0.9977 at twenty minutes. Any error metric computed against it is therefore a measurement of the sampling interval rather than of a model: predicting that the next value equals the current one is almost exactly right, and a model that beats that baseline by a few per cent has demonstrated nothing about the structure. The level is decomposed, because a decomposition of the level is what says what the record is *made of*; it is never scored as a forecast.

4.2 Why forecasting is done on the hourly change

Differencing removes the autocorrelation but introduces a different problem, and the cadence decides which of the two dominates. Table 4 gives both. At twenty minutes the first difference has a lag-one autocorrelation of -0.058 , which is the signature of additive measurement noise on the level rather than of structural memory, and it correlates with the change in air temperature at -0.778 . At one hour the same difference retains $+0.349$ of memory and couples to air temperature at -0.892 . The hourly change is therefore the cadence at which the increment carries signal rather than noise, and it is the target of the forecasting model.

Table 4: The evidence for the two cadences. Read from NP_04_cadence_evidence.csv.

Cadence	Level r_1	Change r_1	Change s.d.	$\rho(\Delta\text{inc}, \Delta\text{tair})$	Drift (mdeg/yr)
20min	0.9977	-0.0581	2.572	-0.7781	-65.97
1h	0.9954	0.3494	3.700	-0.8924	2.30

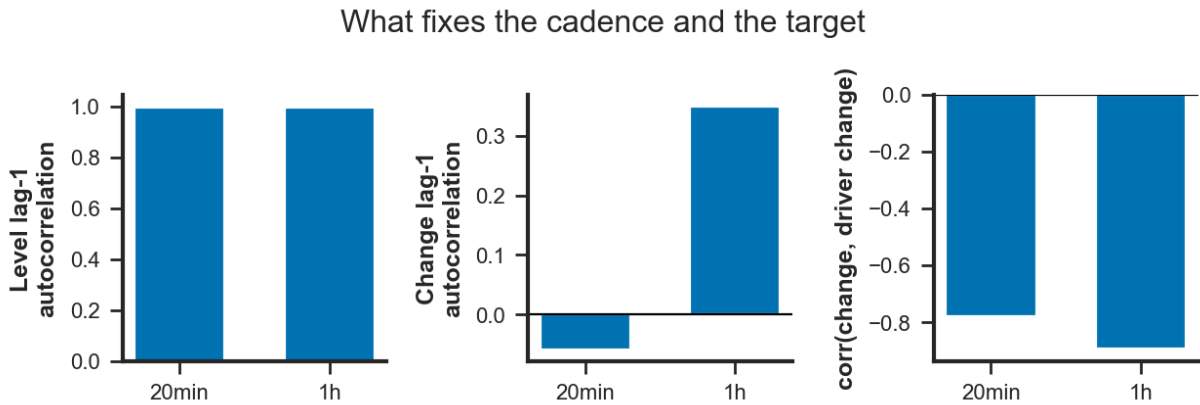


Figure 4: Why the forecasting target is the hourly change rather than the level or the twenty-minute change.

4.3 Why the anomaly test is not differenced, and why contemporaneous regressors are not leakage

The two questions this study puts to the record are different questions, and they are entitled to different rules. Asking *how far ahead can this be forecast* requires that the model see only the past, and the forecasting model of Section 8 is held to that: it consumes lagged predictor values only. Asking *is this reading the expected one* does not. A monitoring system judging a newly

arrived inclination has, in the same acquisition record, the air temperature and relative humidity measured at the same instant — they arrive together, twenty minutes at a time. Consuming them is not leakage but a description of what the system actually holds when it must decide. The two models are never scored on the same table, and no number from one is quoted against the other.

4.4 Why two models, and why the decomposition carries no autoregression

With autoregressive lags enabled, the autoregressive component absorbs most of the diurnal structure and the seasonal component becomes whatever periodicity is left over after it, rather than the wall’s thermal cycle. A decomposition meant to be read as physics must therefore carry no autoregressive term, which is exactly the term a forecast needs. One model cannot do both jobs, so there are two.

4.5 How gaps are honoured

No value is imputed anywhere in this study. Every contiguous run of accepted observations carries its own segment identifier, and the models are told about those identifiers, so that no fitted window and no prediction window spans a gap. This matters because NeuralProphet’s own defaults do not honour it: left to itself the library fills gaps of at least thirty hours, and a decomposition fitted over fabricated data attributes to the structure a shape the structure never produced.

4.6 Changepoints on covered time, and no era offset

The trend’s changepoints are placed at quantiles of the *observed* timestamps rather than uniformly along the axis. A changepoint inside a 103-day outage is constrained by no data and is free to take any value the extrapolation on either side of it requires. Instrument eras are not modelled at all: Study 01 compensated and anchored the record once across the changeover of 21 February 2025, and this study takes that product as delivered. No era offset is estimated and none is subtracted.

4.7 The metrics, and what each is for

Point accuracy is reported as mean absolute error and root mean square error in millidegrees, and as MASE against a naive scale computed on training rows only, so that the scale-free figure is not self-referential. An interval is reported on three counts rather than one, because coverage alone can always be bought by widening: how often it contains the observation, how wide it had to be to manage that, and the Winkler interval score which penalises both together.

5 What the record is made of

Table 5 gives the variance each fitted component carries and the peak-to-peak range it spans. The result is dominated by two things: a trend carrying 48.9% of the variance and spanning 213 mdeg, and a residual carrying 30.4%. Air temperature accounts for 12.2% and the annual cycle 8.4%. The daily seasonality carries almost nothing — 0.03% — and that is a consequence of the model’s construction rather than a finding about the wall: air temperature enters as a contemporaneous regressor, so it absorbs the diurnal variation, and what the daily seasonal term is left to explain is the small residue the temperature does not.

Table 5: Variance share and peak-to-peak range of each fitted component, over the training window. Read from `NP_05_component_shares.csv`.

Component	Variance	Share	Peak-to-peak (mdeg)
trend	1,867.1	48.9%	212.6
residual	1,160.1	30.4%	241.5
future_regressor_tair	464.3	12.2%	108.2
season_yearly	320.7	8.4%	63.6
future_regressor_rh	1.8	0.0%	5.1
season_daily	1.2	0.0%	4.2

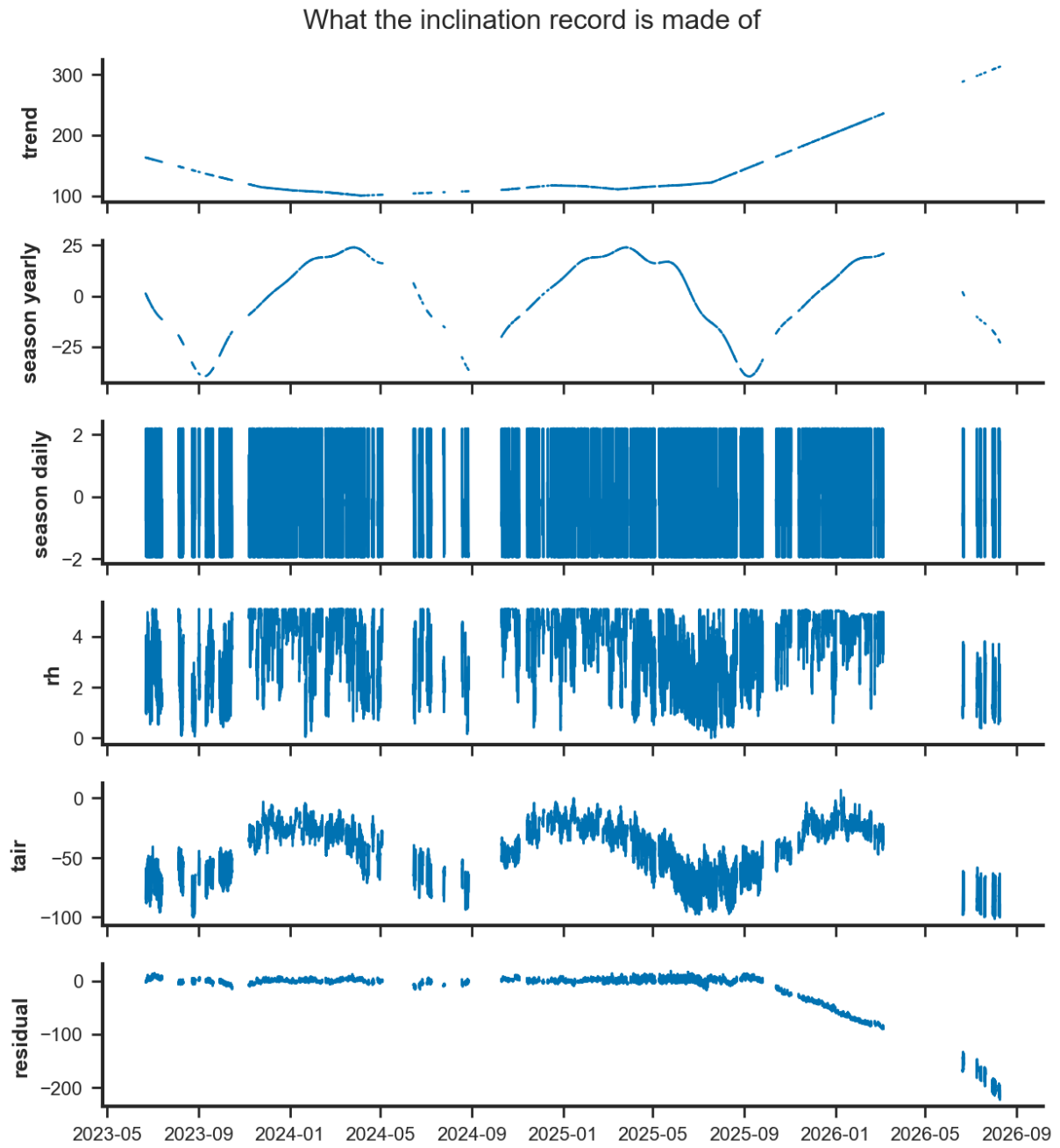


Figure 5: The decomposition of the compensated, cleaned inclination level into trend, annual and daily cycles, the response to the measured environment, and the remainder. Lines break across dropped outages rather than interpolating through them.

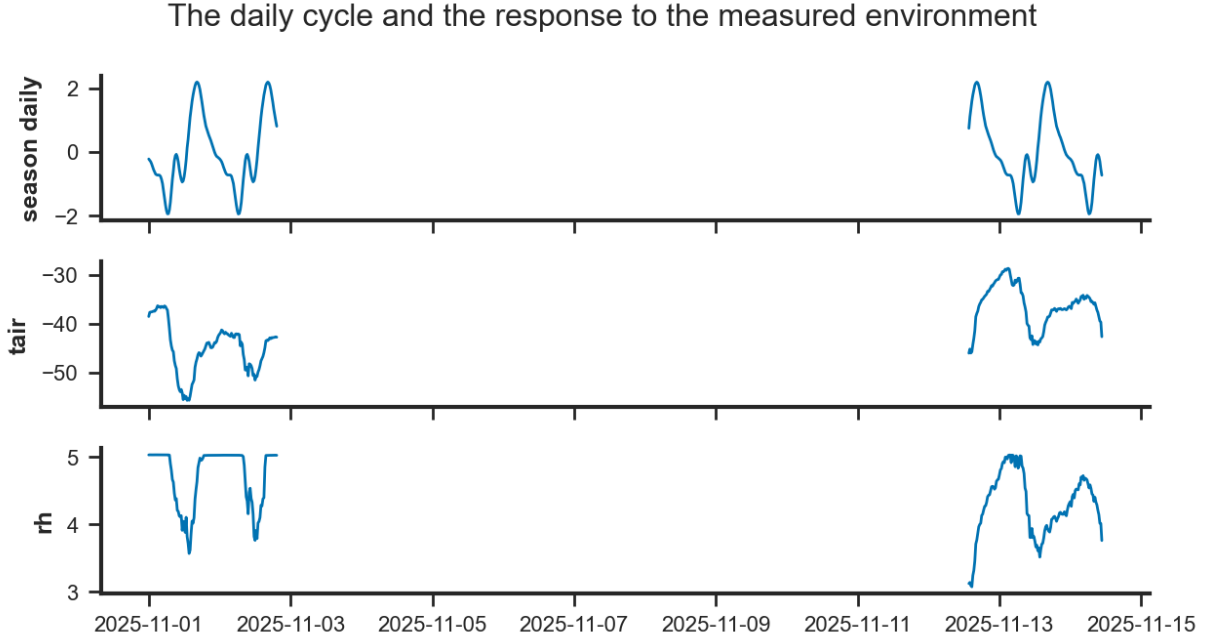


Figure 6: The two interpretable components on their own axes over a fortnight: the daily cycle the model fitted, and the response it attributes to air temperature and relative humidity.

5.1 Confrontation with Study 03

The thermal gain this decomposition learns is an internal quantity of a fitted model, and it is worth only as much as its agreement with an independent measurement. Study 03 measured the diurnal-band gain of the same channel against the same air temperature by a lag-and-gain scan, a method sharing no machinery with this one, and obtained -2.79 mdeg/ $^{\circ}\text{C}$. This decomposition returns -2.56 mdeg/ $^{\circ}\text{C}$. Table 6 places them side by side, together with the two weaker external proxies Study 03 also scanned.

Table 6: The learned thermal gain against Study 03’s independent measurements. Read from NP_06_learned_gains.csv.

Source	Gain (mdeg/ $^{\circ}\text{C}$)	Method
Model A, compensated channel	-2.56	NeuralProphet future regressor
Study 03, diurnal band	-2.79	lag and gain scan
Study 03, ground station	-2.23	lag and gain scan
Study 03, ERA5	-2.04	lag and gain scan

The two agree to within 8%, by methods with nothing in common, and both carry the negative sign the embankment geometry predicts. That is the strongest external check this study has, and it is the reason the decomposition’s thermal term can be read as a description of the wall rather than as a fitting artefact. It is a statement about the *compensated* channel: what the wall does after Study 01’s correction, which is the only quantity a monitoring system ever sees. Whether that correction is itself correct is Study 01’s question and is not reopened here.

5.2 Which components survive re-estimation

A component estimated once is an assertion. Table 7 re-fits the model on each chronological third of the training period and compares the two interpretable quantities across them. The thermal gain is stable: -2.47 , -2.33 and -2.39 mdeg/ $^{\circ}\text{C}$, keeping its sign and its magnitude across three

disjoint stretches. The trend is not: +38.6, +4.8 and +45.5 mdeg/yr, each third disagreeing with the whole-window estimate of -2.77 mdeg/yr in sign as well as size.

Table 7: The two interpretable quantities re-estimated on disjoint thirds of the training period. Read from NP_16_component_stability.csv.

Block	Thermal gain (mdeg/°C)	Trend (mdeg/yr)
third 1	-2.47	+38.6
third 2	-2.33	+4.8
third 3	-2.39	+45.5

The contrast is the finding. The thermal gain behaves like a physical constant of the structure; the trend behaves like an estimate that has not converged. The drift is therefore quoted once, in-sample, at -2.77 mdeg/yr over 21 June 2023 to 1 September 2025, and no extrapolation of it is used anywhere in this study. The monitoring model of Section 6 carries no trend at all for the same reason.

6 Is this reading the one that was expected?

Three expectations were built and scored against each other, and Table 8 reports all three rather than the best. Two are frozen: a model fitted once at the training origin and applied thereafter, in the attribution configuration that carries a trend and in the monitoring configuration that does not. The third is refitted every thirty days on all history available at each origin.

Table 8: The expectation scored three ways. Coverage and width refer to the nominal 90% interval. Read from NP_07_nowcast_metrics.csv.

Fit	n	MAE	RMSE	Bias	MASE	Coverage	Width
rolling, conformal	28,400	18.18	23.68	-10.10	13.10	67.7%	77.89
frozen monitoring	9,920	30.59	31.83	-28.75	22.04	5.8%	10.54
frozen attribution	9,920	58.82	75.18	+57.53	42.39	11.9%	31.11

The refitted expectation wins on every count, and by margins that are not marginal: mean absolute error 18.2 mdeg against 30.6 for the frozen monitoring fit and 58.8 for the frozen attribution fit, bias -10.1 mdeg against -28.8 and $+57.5$, interval score 164 against 511 and 1197. The reason is visible in the bias column. A fit frozen at the training origin sits nearly 29 mdeg away from the record over the following year, and 82% of its mean square error is in that single offset rather than in scatter — it is not noisy, it is stale. Refitting every thirty days holds the expectation to the drift a month can accumulate, about 0.23 mdeg at the measured rate, and it is also what a deployed system would actually do.

The interval is empirical rather than the model’s own. NeuralProphet’s quantile regression fits the residual’s core, and this residual has a tight core with fat tails; its nominal 90% interval covers 68.7% even in-sample. Conformal quantiles, calibrated on residuals from before 1 June 2025 and never on rows the interval is scored against, replace it.

They do not repair it. The reported coverage is **67.7%** against a nominal 90%, at a median width of 77.9 mdeg. This is stated as measured and is not adjusted. Two things account for it, and both are consequences of decisions taken for other reasons. The calibration stretch ends at a fixed date while the minimum training history was raised to a full annual cycle, which shortened that stretch from roughly seventeen months to eleven. And the monitoring model carries no trend by design, so the wall’s own slow movement remains in the residual and continues after the

calibration stretch ends — the interval is calibrated on a quieter record than the one it is then asked to cover. An interval that undercovers by this much is a statement about this expectation, and Section 11 records it as one.

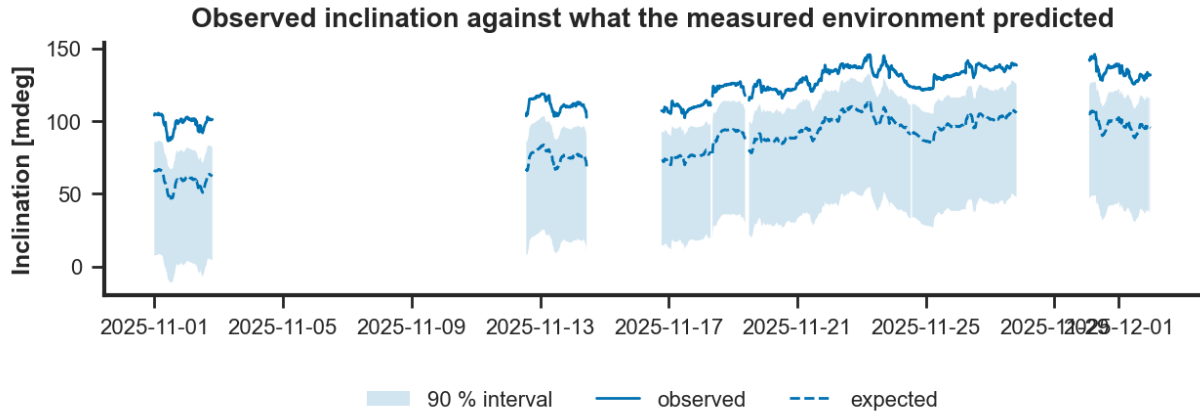


Figure 7: One month of the observed inclination against what the measured environment predicted, with the conformal interval around the expectation.

7 Judging a departure

7.1 The reference window, and what had to be excluded from it

Every control limit below is a multiple of a centre and a scale estimated on a reference window, so a window containing a departure would calibrate the detector against the very thing it is meant to find. Two stretches are therefore excluded, for two different reasons.

The window ends on 1 June 2025, before the stretch Study 01 flagged in summer 2026. That is a precaution about calibration and explicitly not a test: this study makes no claim, anywhere, about whether this detector finds that stretch.

The window also begins later than the monitored record could allow, on 1 October 2024, and this exclusion was forced by measurement rather than anticipated. The residual's thirty-day rolling mean runs -60.5 , -61.0 , -34.2 and -35.5 mdeg over June to September 2024 and then settles to $+3.4$ mdeg and stays within roughly ± 20 mdeg for the remainder of the record. The cause is identifiable: those early refits train on barely one calendar annual cycle, and one calendar year of this record contains outages of 40, 42 and 17 days, so their annual term is fitted to substantially less than one cycle and is not identifiable. Those months are excluded from the monitored record as well as from the reference window. An expectation wrong by 60 mdeg for a reason internal to the model is not a structural departure, and reporting it as one would be a false claim.

7.2 The limit width, and why it is far above the textbook range

The limit is not taken from convention. A false-alarm budget is fixed first — ninety days of watched time per false alarm — and the limit width is then swept until one meets it, so that every detection figure in this study is quoted at a stated budget rather than at whatever rate a conventional setting happens to give.

The width that meets it is 14.75 standard deviations, which is far outside anything a control-chart text would offer, and the reason is a property of this residual that must be stated plainly. EWMA and CUSUM limits are derived for independent observations. This residual has a lag-one autocorrelation of **0.995 at twenty minutes** and still **0.974 at twenty-four hours**. Against a series that correlated, a textbook width between two and five standard deviations does not buy months between false alarms but days: the best any width in that range achieves here is 15.7 days against the ninety asked for. Widening the swept range is not the same as widening the

limit until the alarms stop — the budget was fixed before the sweep, and the width is whatever meets it — but the resulting number should be read as a property of the residual rather than as a conventional control limit.

At 14.75 the achieved run length is 122 days per false alarm. It rests on two episodes over 244 watched days, and with a count that small it bounds the false-alarm rate rather than measuring it. It is quoted here to one significant figure of confidence and no more.

7.3 What the detector then reported

Over the monitored record the joint chart raises 14 alarming episodes covering 350 of 678 watched days. Table 9 gives them. One episode dominates: 288 consecutive days from 24 September 2025, at a mean standardised departure of +5.2.

Table 9: Alarming episodes over the monitored record, where the EWMA and CUSUM charts agree within six hours. Read from `NP_09_alarm_episodes.csv`.

Start	End	Duration (h)	Mean z	Peak $ z $
2025-04-28 08:00	2025-04-28 13:40	6.0	+2.39	3.67
2025-05-02 21:20	2025-05-16 20:40	335.7	+4.97	8.47
2025-06-14 10:00	2025-06-14 23:00	13.3	-2.50	3.33
2025-06-15 07:20	2025-06-16 02:40	19.7	-2.53	3.56
2025-06-16 08:20	2025-06-17 03:40	19.7	-2.44	3.25
2025-06-17 04:40	2025-06-17 10:20	6.0	-2.03	3.37
2025-06-18 10:00	2025-07-15 21:20	659.7	-5.75	10.50
2025-08-09 09:00	2025-08-09 20:20	11.7	-2.43	3.09
2025-08-10 07:40	2025-08-14 20:00	108.7	-2.50	3.63
2025-09-03 17:40	2025-09-04 03:40	10.3	+2.42	3.00
2025-09-04 05:00	2025-09-04 14:40	10.0	+2.23	3.19
2025-09-11 05:00	2025-09-13 16:20	59.7	+2.59	4.18
2025-09-24 19:00	2026-07-09 12:40	6,906.0	+5.20	7.82
2026-07-30 09:40	2026-08-09 15:00	245.7	-4.95	8.56

A detector that spends half its watched time in alarm is reporting something about the model as much as about the wall. The monitoring fit carries no trend by deliberate choice, so the wall’s slow movement is left in the residual where a monitor can see it; the consequence is that when the wall does move slowly and persistently, the chart alarms and keeps alarming. That is the intended behaviour, and it is also why the episode table cannot be read as a list of distinct structural events.

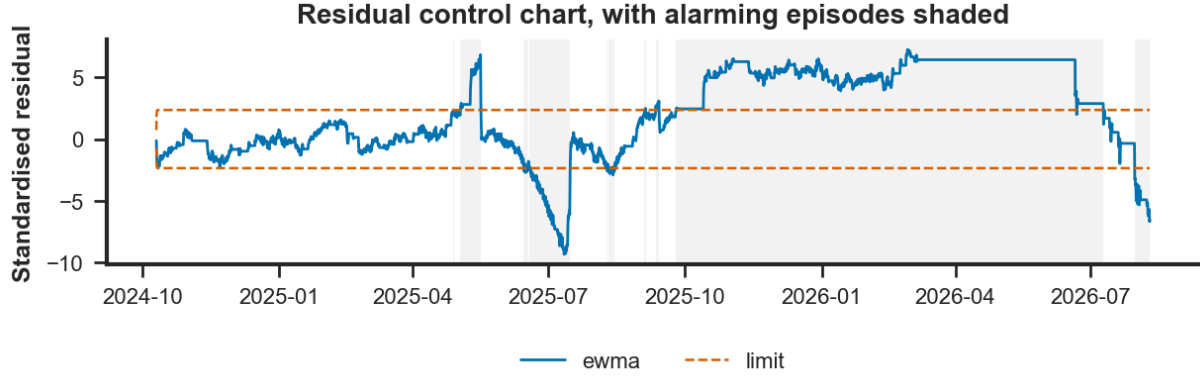


Figure 8: The residual control chart over the monitored record, with alarming episodes shaded. The line breaks across outages rather than interpolating through them.

7.4 What this detector can and cannot see

One large event cannot establish a detector’s sensitivity, and the summer-2026 stretch is not used to try. Departures of known size, duration and *shape* are injected into the uncontaminated residual instead, the charts are re-run, and an alarm counts only where the uncontaminated run is silent within a twenty-four-hour response window.

The three shapes correspond to mechanisms a three-leaf stone wall — two masonry leaves either side of a weaker rubble-and-mortar core — can actually produce. **Amplitude growth:** the leaves stop acting together, through delamination at the core interface or loss of through-stones, and the section bends further under the same daily heating while its timing and mean hold. **Phase change:** the thermal path changes rather than the stiffness, as when water entering the core raises its heat capacity, and the wall answers the same forcing earlier or later. **Drift:** creep of the lime mortar under sustained load, thermal ratcheting of the outer leaf, or foundation settlement — slow, monotone and invisible in any single day.

The result is stark, and it is the most operationally important number in this study. Of the three mechanisms, only amplitude growth is ever detected, and only at 16 mdeg. A timing shift of up to two hours, which the fitted daily amplitude of 2.1 mdeg converts to a residual amplitude of 1.07 mdeg, is not detected at any duration. A drift of up to 20 mdeg per year is not detected at any duration.

Table 10: Detectability by mechanism, magnitude and persistence, at the stated false-alarm budget. Read from NP_10_detectability.csv.

Mechanism	Magnitude	Duration (h)	Detected	Delay (h)
amplitude	0.50	6	no	—
amplitude	0.50	24	no	—
amplitude	0.50	72	no	—
amplitude	0.50	168	no	—
amplitude	0.50	336	no	—
amplitude	1.00	6	no	—
amplitude	1.00	24	no	—
amplitude	1.00	72	no	—
amplitude	1.00	168	no	—
amplitude	1.00	336	no	—
amplitude	2.00	6	no	—
amplitude	2.00	24	no	—
amplitude	2.00	72	no	—
amplitude	2.00	168	no	—
amplitude	2.00	336	no	—
amplitude	4.00	6	no	—
amplitude	4.00	24	no	—
amplitude	4.00	72	no	—
amplitude	4.00	168	no	—
amplitude	4.00	336	no	—
amplitude	8.00	6	no	—
amplitude	8.00	24	no	—
amplitude	8.00	72	no	—
amplitude	8.00	168	no	—
amplitude	8.00	336	no	—
amplitude	16.00	6	yes	7.0
amplitude	16.00	24	yes	32.0
amplitude	16.00	72	yes	79.0
amplitude	16.00	168	no	—
amplitude	16.00	336	yes	248.7
phase	0.14	6	no	—
phase	0.14	24	no	—
phase	0.14	72	no	—
phase	0.14	168	no	—
phase	0.14	336	no	—
phase	0.27	6	no	—
phase	0.27	24	no	—
phase	0.27	72	no	—
phase	0.27	168	no	—
phase	0.27	336	no	—
phase	0.54	6	no	—
phase	0.54	24	no	—
phase	0.54	72	no	—
phase	0.54	168	no	—
phase	0.54	336	no	—
phase	1.07	6	no	—

Two caveats belong with that table. The amplitude field is not monotone: 16 mdeg is detected at 6, 24, 72 and 336 hours but not at 168, which is an artefact of injecting at a single fixed point in the record rather than a property of the mechanism, and it is reported rather than smoothed. And the whole field is quoted at a 122-day false-alarm budget; a system willing to tolerate more false alarms would see smaller departures, and the trade is the operator's to make.

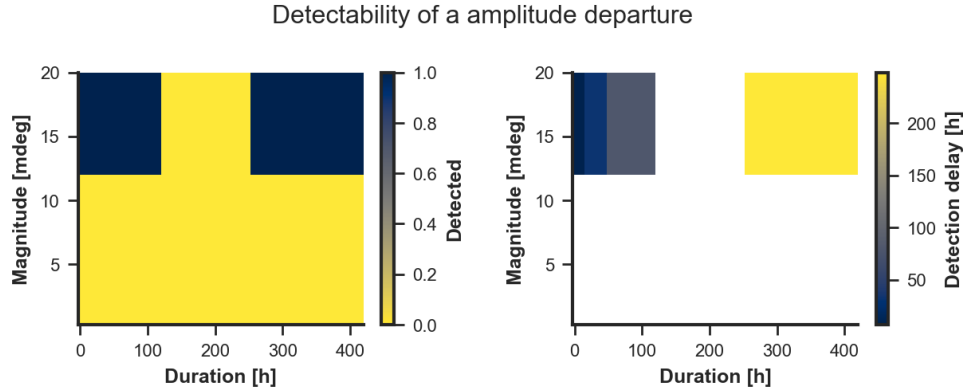


Figure 9: Detectability of a growth in the daily swing, the signature of lost composite action between the leaves.

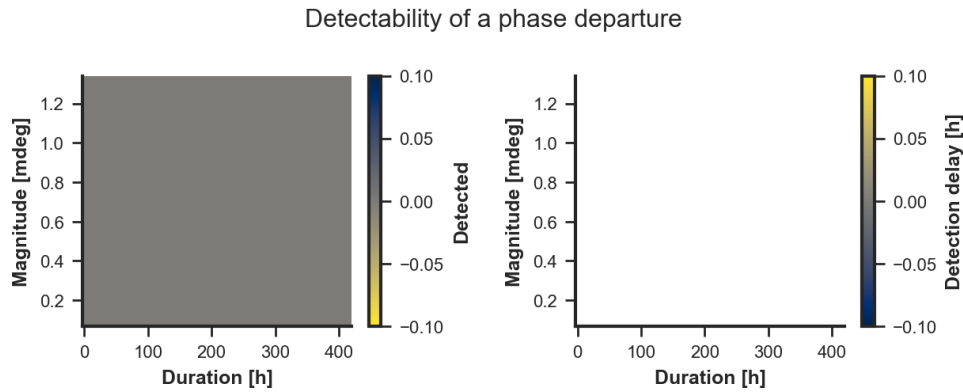


Figure 10: Detectability of a timing shift, the signature of a changed thermal path such as water in the core. Nothing in the swept range is found.

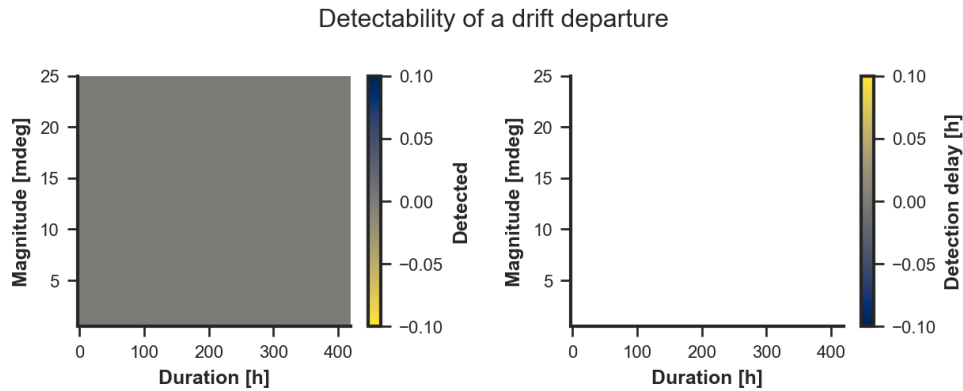


Figure 11: Detectability of a slow monotone drift, the signature of mortar creep or settlement. Nothing in the swept range is found.

8 How far ahead is prediction worth anything?

The target here is the gap-safe one-hour change, formed only between adjacent accepted observations inside one contiguous run, so that no difference bridges a gap. After segmentation at a minimum run length of 96 hours, 14,930 hourly rows in 59 runs remain, evaluated over five expanding whole-segment folds.

Table 11 gives the error of each model and each baseline against horizon. Every model reaches a mean absolute error between 1.77 and 1.84 mdeg at every horizon from one hour to 48; the zero-change baseline sits at 2.17 mdeg, seasonal-naive at 2.24, and persistence between 2.53 and 3.62.

Table 11: Forecast error by model and horizon, against three baselines a monitoring system could run for free. Read from NP_11_forecast_metrics.csv.

Model	Horizon (h)	n	MAE	RMSE	MASE	Coverage
AR + batt (control)	1	7,802	1.817	2.782	0.873	81.0%
AR + batt (control)	3	7,802	1.785	2.703	0.857	81.0%
AR + batt (control)	6	7,802	1.785	2.712	0.857	80.9%
AR + batt (control)	12	7,802	1.774	2.697	0.852	81.8%
AR + batt (control)	24	7,802	1.803	2.739	0.866	81.2%
AR + batt (control)	48	7,802	1.841	2.822	0.884	82.1%
AR + tair	1	7,802	1.810	2.782	0.869	77.0%
AR + tair	3	7,802	1.778	2.708	0.854	77.5%
AR + tair	6	7,802	1.774	2.708	0.852	77.8%
AR + tair	12	7,802	1.773	2.707	0.852	77.9%
AR + tair	24	7,802	1.801	2.743	0.865	77.7%
AR + tair	48	7,802	1.838	2.826	0.883	78.7%
AR + tair + rh	1	7,802	1.768	2.741	0.849	77.4%
AR + tair + rh	3	7,802	1.746	2.689	0.839	77.7%
AR + tair + rh	6	7,802	1.753	2.707	0.842	78.0%
AR + tair + rh	12	7,802	1.768	2.713	0.849	78.0%
AR + tair + rh	24	7,802	1.787	2.738	0.858	77.7%
AR + tair + rh	48	7,802	1.829	2.829	0.879	78.1%
AR only	1	7,802	1.816	2.783	0.872	75.6%
AR only	3	7,802	1.783	2.704	0.856	75.9%
AR only	6	7,802	1.780	2.703	0.855	76.5%
AR only	12	7,802	1.774	2.702	0.852	76.4%
AR only	24	7,802	1.803	2.740	0.866	76.2%
AR only	48	7,802	1.841	2.826	0.884	76.6%
persistence	1	10,532	2.527	4.008	1.214	—
persistence	3	10,454	2.955	4.523	1.419	—
persistence	6	10,337	3.414	5.167	1.640	—
persistence	12	10,103	3.621	5.403	1.739	—
persistence	24	9,635	2.244	3.346	1.078	—
persistence	48	8,699	2.318	3.461	1.114	—
seasonal_naive	1	9,635	2.244	3.346	1.078	—
seasonal_naive	3	9,635	2.244	3.346	1.078	—
seasonal_naive	6	9,635	2.244	3.346	1.078	—
seasonal_naive	12	9,635	2.244	3.346	1.078	—
seasonal_naive	24	9,635	2.244	3.346	1.078	—
seasonal_naive	48	8,699	2.318	3.461	1.114	—
zero	1	10,571	2.172	3.471	1.043	—
zero	3	10,571	2.172	3.471	1.043	—
zero	6	10,571	2.172	3.471	1.043	—
zero	12	10,571	2.172	3.471	1.043	—
zero	24	10,571	2.172	3.471	1.043	—
zero	48	10,571	2.172	3.471	1.043	—

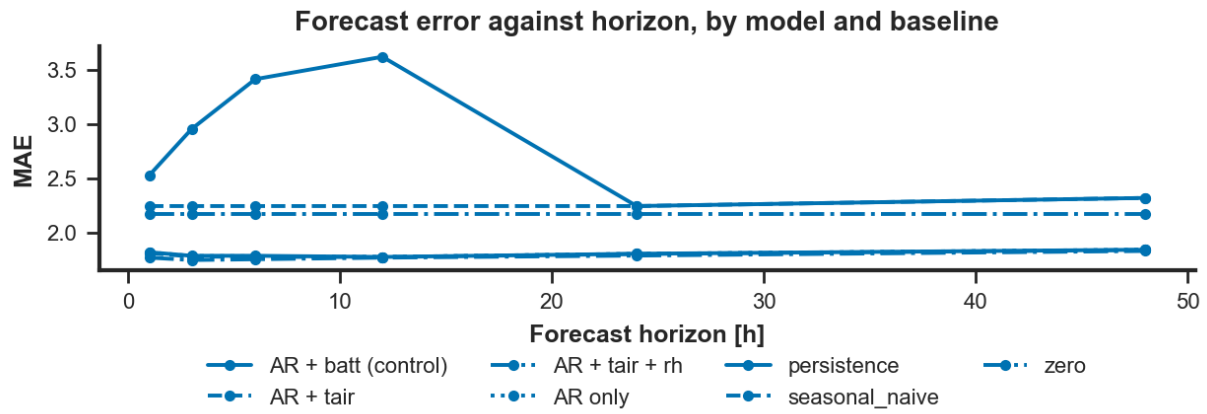


Figure 12: Forecast error against horizon, by model and baseline.

An error is not a result until it is compared, and a comparison is not evidence until it is paired. Table 12 gives the fractional reduction in mean absolute error against each baseline, computed on the timestamps the two share and with a block bootstrap whose block length is adapted to the horizon, because these residuals are autocorrelated and an unpaired comparison would report a significance that is not there.

Table 12: Paired block-bootstrap skill against each baseline, with 5th and 95th percentiles of the bootstrap distribution. Read from NP_12_skill_vs_baseline.csv.

Baseline	Model	Horizon (h)	Skill	q_{05}	q_{95}
zero	AR only	1	+0.160	+0.144	+0.176
zero	AR only	3	+0.177	+0.159	+0.194
zero	AR only	6	+0.181	+0.164	+0.198
zero	AR only	12	+0.187	+0.170	+0.204
zero	AR only	24	+0.172	+0.152	+0.192
zero	AR only	48	+0.160	+0.135	+0.184
zero	AR + tair	1	+0.164	+0.147	+0.179
zero	AR + tair	3	+0.179	+0.162	+0.196
zero	AR + tair	6	+0.184	+0.167	+0.200
zero	AR + tair	12	+0.187	+0.170	+0.204
zero	AR + tair	24	+0.173	+0.153	+0.193
zero	AR + tair	48	+0.161	+0.137	+0.185
zero	AR + tair + rh	1	+0.183	+0.167	+0.198
zero	AR + tair + rh	3	+0.194	+0.178	+0.209
zero	AR + tair + rh	6	+0.194	+0.178	+0.210
zero	AR + tair + rh	12	+0.190	+0.173	+0.206
zero	AR + tair + rh	24	+0.180	+0.160	+0.199
zero	AR + tair + rh	48	+0.165	+0.142	+0.189
zero	AR + batt (control)	1	+0.160	+0.144	+0.176
zero	AR + batt (control)	3	+0.176	+0.159	+0.193
zero	AR + batt (control)	6	+0.179	+0.162	+0.196
zero	AR + batt (control)	12	+0.187	+0.170	+0.204
zero	AR + batt (control)	24	+0.172	+0.152	+0.192
zero	AR + batt (control)	48	+0.160	+0.134	+0.185
persistence	AR only	1	+0.274	+0.265	+0.282
persistence	AR only	3	+0.397	+0.386	+0.407
persistence	AR only	6	+0.480	+0.468	+0.493
persistence	AR only	12	+0.510	+0.496	+0.523
persistence	AR only	24	+0.189	+0.175	+0.202
persistence	AR only	48	+0.203	+0.187	+0.220
persistence	AR + tair	1	+0.277	+0.268	+0.284
persistence	AR + tair	3	+0.398	+0.388	+0.409
persistence	AR + tair	6	+0.482	+0.470	+0.494
persistence	AR + tair	12	+0.510	+0.496	+0.524
persistence	AR + tair	24	+0.189	+0.175	+0.203
persistence	AR + tair	48	+0.204	+0.187	+0.221
persistence	AR + tair + rh	1	+0.293	+0.285	+0.301
persistence	AR + tair + rh	3	+0.409	+0.399	+0.419
persistence	AR + tair + rh	6	+0.488	+0.477	+0.500
persistence	AR + tair + rh	12	+0.512	+0.498	+0.525
persistence	AR + tair + rh	24	+0.196	+0.181	+0.210
persistence	AR + tair + rh	48	+0.208	+0.190	+0.226
persistence	AR + batt (control)	1	+0.273	+0.265	+0.281
persistence	AR + batt (control)	3	+0.396	+0.386	+0.407
persistence	AR + batt (control)	6	+0.479	+0.467	+0.491
persistence	AR + batt (control)	12	+0.510	+0.496	+0.523

The answer to the third question is therefore that *the horizon at which forecasting stops beating a trivial baseline was not reached within the range tested*. Skill against persistence runs from 0.277 at one hour to 0.204 at 48 hours, and against the zero-change baseline from 0.164 to 0.161; every bootstrap interval excludes zero, including at the longest horizon. The decay with horizon is mild because the quantity being forecast is a one-hour increment, not a level: the baselines degrade as fast as the model does.

8.1 What each predictor actually bought

A model carrying both autoregressive memory and air temperature cannot say which of the two earned its skill, which is why the ladder exists. Its result is the most useful negative finding in this study.

Table 13: The skill increment each predictor bought over the rung below it. Read from NP_13_ablation.csv.

Over	Model	Horizon (h)	Skill	q_{05}	q_{95}
AR only	AR + tair	1	+0.004	+0.003	+0.005
AR only	AR + tair	3	+0.003	+0.001	+0.004
AR only	AR + tair	6	+0.003	+0.001	+0.005
AR only	AR + tair	12	+0.000	-0.001	+0.002
AR only	AR + tair	24	+0.001	+0.000	+0.002
AR only	AR + tair	48	+0.001	-0.001	+0.003
AR + tair	AR + tair + rh	1	+0.023	+0.019	+0.027
AR + tair	AR + tair + rh	3	+0.018	+0.015	+0.021
AR + tair	AR + tair + rh	6	+0.012	+0.009	+0.015
AR + tair	AR + tair + rh	12	+0.003	+0.001	+0.005
AR + tair	AR + tair + rh	24	+0.008	+0.006	+0.010
AR + tair	AR + tair + rh	48	+0.005	+0.001	+0.009

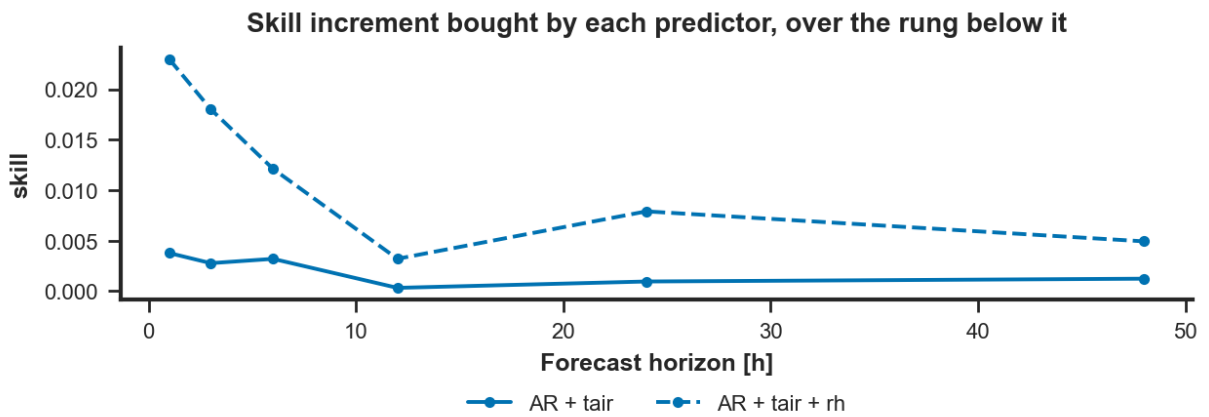


Figure 13: The skill increment each predictor bought over the rung below it. Note the vertical scale: both increments are small fractions of a per cent to a few per cent, against the 16 to 28% the models take from the baselines.

Air temperature, added to autoregression alone, buys 0.0038 of skill at one hour and less than 0.0013 at every horizon beyond six; at 12 and 48 hours its bootstrap interval includes zero.

Relative humidity, added on top of air temperature, buys rather more — 0.023 at one hour, with an interval excluding zero at every horizon — though the magnitudes remain small.

The negative control settles the interpretation. Battery voltage added to autoregression scores 0.273 against persistence at one hour, where autoregression alone scores 0.274. Adding a channel to this model buys nothing by itself, so the 0.0038 that air temperature buys is a real increment and not an artefact of an extra input; it is simply a very small one.

This does not contradict Section 5. The thermal coupling is real, was measured here at -2.56 mdeg/ $^{\circ}\text{C}$ and independently by Study 03 at -2.79 , and it explains 12.2% of the level's variance. It is nonetheless almost useless for *forecasting the change*, because the response's own recent history already carries the information the temperature would supply. A coupling that is real is not obliged to be predictively useful, and the distinction between the two matters for anyone deciding which sensors a deployment needs: this record argues for keeping the inclinometer well sampled rather than for adding environmental channels.

9 Can the model fill the gaps it was trained around?

The operational temptation, once a model predicts a change, is to accumulate its predictions across a gap and call the result a reconstructed level. That is a stronger claim than anything measured above: a per-step error that is small and unbiased still accumulates, and a reconstruction arriving at the wrong level on the far side of a gap is worse than an admitted absence, because it looks like a measurement.

The check was to be arithmetical. Where a gap is bracketed by accepted observations on both sides the true change across it is known, so the predicted changes can be summed and compared, and a reconstruction is safe only if it closes.

The question is undecidable on this pipeline, and the reason is structural rather than numerical. Of the 391 hourly gaps, 390 return `incomplete_estimates` and one `unbracketed`; none is scorable. The forecasting model produces predictions only at timestamps that survive segmentation, and those are by construction the timestamps at which an observation exists. It therefore never predicts *inside* a gap at all, and there is nothing to accumulate. Answering the question properly requires rolling the model forward recursively through each gap, feeding each prediction back as the next step's history, which this pipeline does not do.

Table 14: Gap closure summarised by status, which is what decides whether a gap can be scored at all. None of the 391 gaps is scorable. The full per-gap table is in `NP_14_gap_closure.csv`.

Status	Gaps	Missing slots	Median error
<code>incomplete_estimates</code>	390	6,008	—
<code>unbracketed</code>	1	14	—

The honest verdict is therefore that this study does not license gap filling by accumulation, on the ground that it has not tested it — which is a weaker statement than "unsafe" and a much weaker one than "safe". Section 11 records it as the study's largest piece of unfinished business.

10 Verdict

What is the record made of? A trend carrying 48.9% of the variance and spanning 213 mdeg, a residual carrying 30.4%, the response to air temperature 12.2% and the annual cycle 8.4%. The thermal gain is -2.56 mdeg/ $^{\circ}\text{C}$, which agrees within 8% with Study 03's independently measured -2.79 and carries the sign the embankment geometry predicts. That gain survives re-estimation on three disjoint stretches; the trend does not, disagreeing with itself in sign as well as magnitude, so the drift is quoted once in-sample at -2.77 mdeg/yr and never extrapolated.

Is this reading the expected one? Yes, to within an expectation whose mean absolute error is 18.2 mdeg, provided the model is refitted on a schedule. A frozen fit is not noisy but stale — 82% of its error is a single offset — and refitting every thirty days removes that. The interval around that expectation is the weak point: its conformal 90% band covers 67.7%, because the drift the monitoring model deliberately leaves in the residual keeps moving after the calibration stretch ends. A reading outside that band is evidence; a reading inside it is weaker evidence than the nominal figure would suggest.

How far ahead is prediction worth anything? Further than this study tested. Every model beats every baseline out to 48 hours with bootstrap intervals excluding zero, at a skill of 0.20 against persistence and 0.16 against zero-change. But almost all of that skill is the response’s own memory: air temperature adds 0.4% at one hour and nothing distinguishable from zero beyond six, and a battery-voltage control scores the same as autoregression alone. The environmental channels earn their place in the *decomposition*, not in the *forecast*.

And what can the monitor see? At a measured false-alarm budget of 122 days, only a growth of 16 mdeg in the daily swing. A timing shift of up to two hours and a drift of up to 20 mdeg per year pass undetected across the whole range swept. This is the number an operator would need first, and it is sobering: the configuration tested here detects gross change and not the early, slow signatures that make monitoring worthwhile.

11 Limitations

The compensation is not independent of the thermal term being fitted. The channel modelled here has already been temperature-compensated by Study 01, and this study fits a thermal response to it. The agreement with Study 03 is reassuring because that measurement was made on the same compensated channel by a different method, but neither study can rule out that the compensation and the fitted term are partly describing the same correction.

Two channels were excluded and cover 17% of the window. Wall temperature and solar radiation are the two variables most likely to explain the residual this study cannot account for, and they were set aside because the instruments recording them did not exist for three fifths of the window.

The battery control is not a silent channel. Study 03 measured $r = -0.673$ for supply voltage in the diurnal band, so it marks a conservative floor rather than a zero. A predictor is credited here only when it clears that floor, which makes the ablation’s negative result stronger, not weaker.

Three annual cycles with a 103-day hole cannot identify a yearly term. The window spans 3.2 annual cycles, the last of them interrupted for 103 days. The annual component is retained because leaving a known cause unmodelled would push it into the residual where a detector would read it as structure, but it is not well identified, and the burn-in excluded from the reference window in Section 7 is a direct symptom of that.

The control limit is far outside its textbook range, and rests on two episodes. A limit of 14.75 standard deviations is not a conventional control limit, and it is wide because the residual has a lag-one autocorrelation of 0.995 — a condition under which EWMA and CUSUM theory does not hold. The achieved run length of 122 days rests on two excursions and bounds the false-alarm rate rather than measuring it. Two alternative designs were not tested and would each be a better answer than a wider limit: charting prewhitened innovations, and charting a coarser aggregate.

Gap filling was not tested at all, for the structural reason given in Section 9. Nothing here licenses reconstruction by accumulation.

One sensor, one station, one site. Every number above describes the inclinometer at station 02 on the wall at Porta di Sant’Ubaldo. The method is intended to transfer; none of these values is.

The summer-2026 stretch is not evidence of detection. It was excluded from the reference window as a calibration precaution, and no claim is made anywhere in this report about whether the detector finds it. A single large event cannot establish a detection rate, and finding it would show only that the detector is not broken.

12 Run metadata

Every parameter and library version behind the artefacts quoted above.

Table 15: Run metadata. Read from NP_15_run_metadata.csv.

Parameter	Value
segment_start	2023-06-21
native_freq	20min
model_a_freq	20min
model_b_freq	1h
target_column	inc_comp_cleaned
predictors	tair, rh
model_a_lags	0
model_a_growth	linear
model_a_monitor_growth	off
model_a_changepoints	12
model_a_yearly	True
model_a_train_end	2025-09-01
model_a_epochs	30
rolling_refit_every	30d
rolling_min_train	365d
conformal_calibration_end	2025-06-01
model_b_lags	24
model_b_forecasts	48
model_b_regressor_lags	12
model_b_folds	5
model_b_refit_each_fold	False
model_b_epochs	30
seed	0
quantiles	(0.05, 0.95)
reference_window	2024-10-01 to 2025-06-01
ewma_lambda	0.05
ewma_L	14.75
cusum_k	0.5
cusum_h	5.0
joint_window	6h
target_arl_days	90.0
achieved_arl_days	122.0
gap_closure_tolerance_mdeg	1.0
neuralprophet_version	0.8.0
pandas_version	2.3.3
numpy_version	1.26.4